Antibiotic Resistance Prediction Using Genomic Features and Deep Learning

A system that:

1. Accepts bacterial genomic data (FASTA/FASTQ)
2. Extracts **key genomic features**, e.g. Presence/absence of known resistance genes, Mutations in antibiotic-targeted genes, Plasmid presence
3. Encodes these features as input vectors
4. Trains a **deep learning model** (e.g., feedforward neural network, CNN) to predict: Resistant vs susceptible for each antibiotic

Goal: **Predict effective antibiotics before treatment** → save lives, reduce antibiotic misuse.

Convert genomes into **numerical feature vectors**:

- **Binary gene presence matrix**:
  `1` if resistance gene exists, `0` if absent
- **Mutation counts**: Number of known point mutations in key genes
- **Plasmid indicators**: `1` if plasmid conferring resistance exists
- **K-mer counts** (optional, sequence subsequences used as features)

Advance version (Gene-Gene relation)

1. Input: Bacterial Genomic Data

- The system takes **bacterial genome sequences** (FASTA/FASTQ files) as input.
- These genomes can be from publicly available databases or hospital samples.
- From the genome, the system identifies:
  - **Resistance genes** (genes known to confer resistance)
  - **Mutations** in critical antibiotic target genes
  - **Plasmids** carrying resistance genes

This forms the **feature set** for prediction

2. Modeling Gene Interactions with Graph Neural Networks (GNNs)

Instead of treating genes independently, the system models **gene-gene relationships** as a **graph**:

- **Nodes:** Genes or plasmids

- **Edges:** Known biological interactions or co-occurrence patterns
- **Node features:** Mutations, presence/absence of resistance genes

A **GNN** (e.g., Graph Convolutional Network or Graph Attention Network) learns patterns of resistance based on these interactions. This allows the model to understand **complex dependencies** that simple ML models might miss, such as how multiple genes together can confer resistance.

3. Predicting Antibiotic Resistance

- For each antibiotic, the model predicts: Probability that the bacteria are resistant and Probability that the antibiotic will be effective
- This gives a **personalized resistance profile** for the infection.

4. Suggesting Antibiotics Likely to Succeed

- Based on predicted resistance probabilities, the system **recommends antibiotics most likely to succeed**.
- Can support doctors in **choosing effective treatment immediately**, avoiding trial-and-error.
- Reduces Delay in effective therapy, Misuse of broad-spectrum antibiotics and Spread of resistant bacteria

Term	Meaning (One-liner)	Use in Your Project
Bacteria	Single-celled organisms, some cause infections	Each genome you analyze comes from a bacterial sample for which you predict antibiotic resistance
Antibiotics	Drugs that kill or stop bacterial growth	The target drugs for which your model predicts resistance or susceptibility
Antibiotic Resistance	When bacteria survive antibiotic treatment	The main outcome your model predicts (resistant or susceptible)
Genome	Complete DNA sequence of a bacterium	Input data for your model; features are extracted from it
Gene	DNA segment coding for a protein/function	Identify resistance genes as features for prediction
Resistance Gene	Gene that allows bacteria to survive antibiotics	Presence/absence forms binary input features for the model
Mutation	Change in DNA sequence of a gene	Mutations in key genes are extracted as features for prediction
MIC (Minimum Inhibitory Concentration)	Minimum drug amount to stop bacterial growth	Used as label/ground truth to define resistant vs susceptible
Plasmid	Extra DNA in bacteria carrying genes	Detect plasmids carrying resistance genes to enhance feature set
K-mer	Short DNA subsequence	Optional input feature representing sequence patterns for deep learning